Project #3: SSD Performance Profiling

Modern SSDs, especially NVMe SSDs, can deliver very high IOPS and throughput when requests are issued concurrently. Like memory, storage exhibits a throughput-latency tradeoff governed by queueing: increasing queue depth improves utilization until a saturation knee, after which latency rises sharply with little additional throughput.

Data-loss Warning
Never benchmark against a partition that contains valuable data. Use a dedicated empty partition, raw device reserved for tests, or a large disposable file with direct I/O. You are responsible for preventing data loss.

Learning Goals

- Measure QD=1 latency for reads and writes under random 4KiB and sequential 128KiB patterns.
- Measure maximum small-block IOPS and large-block sequential throughput.
- Reveal throughput-latency behavior as queue depth and/or parallelism increases.
- Explain block-size, access-pattern, read/write mix, working-set/LBA-range, and device-state effects.
- Characterize p95/p99/p99.9 tail latency and relate it to queueing and device state.

Tools

- FIO for controlled storage workloads: block size, queue depth, numjobs, direct I/O, patterns, mixes, and percentiles.
- Optional: OS disk statistics and SMART/health tools for temperature, throttling, and write-count observations.

Experimental Knobs

- Block size: 4KiB, 16KiB, 32KiB, 64KiB, 128KiB, 256KiB; optionally 512KiB or 1MiB for sequential.
- Access pattern: sequential vs. random; maintain alignment for random tests.
- Read/write ratio: 100% read, 100% write, 70/30, and 50/50.
- Queue depth and parallelism: sweep iodepth and/or numjobs, e.g., 1 to 256 as device supports.
- Data pattern: incompressible vs. compressible, held consistent and documented.
- Device state: fresh/burst vs. steady-state, temperature, SLC cache exhaustion, and background garbage collection.

Required Experiments and Plots

- Zero-queue baselines: QD=1 latency for 4KiB random read/write and 128KiB sequential read/write; include p95/p99.
- Block-size sweep: throughput and latency from the same runs for random and sequential patterns.
- Read/write mix sweep under fixed block size and pattern.
- Queue-depth/parallelism sweep: single throughput-vs-latency curve, knee identification, and Little's Law discussion.
- Tail-latency characterization at mid-QD and near-knee QD.
- Working-set/LBA-range and device-state note: discuss burst vs. steady-state if observable.
- Demonstrate or explain one misleading setup, such as buffered I/O/page-cache effects, wrong effective queue depth, insufficient preconditioning, or thermal throttling.

Deliverables

- FIO configs, scripts, raw results, parsed data, and plotting code.
- Methodology: SSD model, interface, filesystem vs. raw device, direct I/O usage, capacity/range used, OS/kernel, CPU, temperature if available.
- Plots/tables with units, error bars when feasible, and clear labels for fixed vs. varied knobs.
- Analysis grounded in queueing theory, controller behavior, interface ceilings, and tail-latency implications.

Grading Rubric

Total: 185 points

Category	Points	Description
Safety and zero-queue baselines	30	Safe setup, direct I/O/alignment, and accurate QD=1 latency table.
Block-size and pattern sweeps	35	Complete matrix and clear IOPS/MB/s vs latency interpretation.
Read/write and queue-depth analysis	40	Correct mix and QD experiments, knee identification, Little's Law reasoning.
Tail latency and device state	25	p50/p95/p99/p99.9 reporting and interpretation of queueing/thermal/GC effects.
Reproducibility and data hygiene	25	Configs, raw data, scripts, repeated trials, and environment documentation.
AI-use transparency and pitfall analysis	15	Honest AI-use note and meaningful misleading-setup analysis.
Report quality	15	Clear figures, concise explanation, and thoughtful limitations.